Trajectory_Analysis_BM

June 1, 2026

[ ]:
```
!pip install scanpy
!pip install anndata
!pip3 install igraph
!pip install celltypist
!pip install decoupler
!pip install fa2-modified
!pip install louvain
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Collecting scanpy
  Downloading scanpy-1.11.5-py3-none-any.whl.metadata (9.3 kB)
Collecting anndata>=0.8 (from scanpy)
  Downloading anndata-0.12.6-py3-none-any.whl.metadata (10.0 kB)
Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (3.15.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages
(from scanpy) (1.5.2)
Collecting legacy-api-wrap>=1.4.1 (from scanpy)
  Downloading legacy_api_wrap-1.5-py3-none-any.whl.metadata (2.2 kB)
Requirement already satisfied: matplotlib>=3.7.5 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (3.10.0)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from scanpy) (8.4.0)
Requirement already satisfied: networkx>=2.7.1 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (3.5)
Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.60.0)
Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (2.0.2)
Requirement already satisfied: packaging>=21.3 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (25.0)
Requirement already satisfied: pandas>=1.5.3 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (2.2.2)
Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (1.0.2)
Requirement already satisfied: pynndescent>=0.5.13 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.13)
Requirement already satisfied: scikit-learn>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (1.6.1)
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Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (1.16.3)
Requirement already satisfied: seaborn>=0.13.2 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.13.2)
Collecting session-info2 (from scanpy)
  Downloading session_info2-0.2.3-py3-none-any.whl.metadata (3.4 kB)
Requirement already satisfied: statsmodels>=0.14.5 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.14.5)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from scanpy) (4.67.1)
Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)
Requirement already satisfied: umap-learn>=0.5.6 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.9.post2)
Collecting array-api-compat>=1.7.1 (from anndata>=0.8->scanpy)
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Collecting zarr!=3.0.*,>=2.18.7 (from anndata>=0.8->scanpy)
  Downloading zarr-3.1.5-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy)
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Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba!=0.62.0rc1,>=0.57.1->scanpy)
(0.43.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.5.3->scanpy) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.5.3->scanpy) (2025.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.3->scanpy)
(3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.7.5->scanpy) (1.17.0)
Collecting donfig>=0.8 (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
  Downloading donfig-0.8.1.post1-py3-none-any.whl.metadata (5.0 kB)
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Requirement already satisfied: google-crc32c>=1.5 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (1.7.1)
Collecting numcodecs>=0.14 (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
  Downloading numcodecs-0.16.5-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata
(3.4 kB)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (6.0.3)
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34.9 MB/s eta 0:00:00
Downloading anndata-0.12.6-py3-none-any.whl (172 kB)
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9.1 MB/s eta 0:00:00
Downloading legacy_api_wrap-1.5-py3-none-any.whl (10 kB)
Downloading session_info2-0.2.3-py3-none-any.whl (16 kB)
Downloading array_api_compat-1.12.0-py3-none-any.whl (58 kB)
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Downloading zarr-3.1.5-py3-none-any.whl (284 kB)
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Downloading donfig-0.8.1.post1-py3-none-any.whl (21 kB)
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manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (9.2 MB)
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45.2 MB/s eta 0:00:00
Installing collected packages: session-info2, numcodecs, legacy-api-wrap,
donfig, array-api-compat, zarr, anndata, scanpy
Successfully installed anndata-0.12.6 array-api-compat-1.12.0 donfig-0.8.1.post1
legacy-api-wrap-1.5 numcodecs-0.16.5 scanpy-1.11.5 session-info2-0.2.3
zarr-3.1.5
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata) (1.5)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata) (8.4.0)
Requirement already satisfied: numpy>=1.26 in /usr/local/lib/python3.12/dist-
packages (from anndata) (2.0.2)
Requirement already satisfied: packaging>=24.2 in
/usr/local/lib/python3.12/dist-packages (from anndata) (25.0)
Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in
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/usr/local/lib/python3.12/dist-packages (from anndata) (2.2.2)
Requirement already satisfied: scipy>=1.12 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
/usr/local/lib/python3.12/dist-packages (from anndata) (3.1.5)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: google-crc32c>=1.5 in
/usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata)
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Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->pandas!=2.1.2,>=2.1.0->anndata) (1.17.0)
Collecting igraph
  Downloading igraph-1.0.0-cp39-abi3-manylinux_2_28_x86_64.whl.metadata (4.4 kB)
Collecting texttable>=1.6.2 (from igraph)
  Downloading texttable-1.7.0-py2.py3-none-any.whl.metadata (9.8 kB)
Downloading igraph-1.0.0-cp39-abi3-manylinux_2_28_x86_64.whl (5.7 MB)
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101.6 MB/s eta 0:00:00
Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)
Installing collected packages: texttable, igraph
Successfully installed igraph-1.0.0 texttable-1.7.0
Collecting celltypist
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                           45.1/45.1 kB
3.1 MB/s eta 0:00:00
Requirement already satisfied: numpy>=1.19.0 in
/usr/local/lib/python3.12/dist-packages (from celltypist) (2.0.2)
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/usr/local/lib/python3.12/dist-packages (from celltypist) (1.6.1)
Requirement already satisfied: openpyxl>=3.0.4 in
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/usr/local/lib/python3.12/dist-packages (from celltypist) (3.1.5)
Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.12/dist-
packages (from celltypist) (8.3.1)
Requirement already satisfied: requests>=2.23.0 in
/usr/local/lib/python3.12/dist-packages (from celltypist) (2.32.4)
Requirement already satisfied: scanpy>=1.7.0 in /usr/local/lib/python3.12/dist-
packages (from celltypist) (1.11.5)
Collecting leidenalg>=0.9.0 (from celltypist)
  Downloading leidenalg-0.11.0-cp38-abi3-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (10 kB)
Requirement already satisfied: igraph<2.0,>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from leidenalg>=0.9.0->celltypist)
(1.0.0)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-
packages (from openpyxl>=3.0.4->celltypist) (2.0.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist)
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Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.0.5->celltypist) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.0.5->celltypist) (2025.2)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist)
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Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests>=2.23.0->celltypist) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist)
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Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist)
(2025.11.12)
Requirement already satisfied: anndata>=0.8 in /usr/local/lib/python3.12/dist-
packages (from scanpy>=1.7.0->celltypist) (0.12.6)
Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-
packages (from scanpy>=1.7.0->celltypist) (3.15.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages
(from scanpy>=1.7.0->celltypist) (1.5.2)
Requirement already satisfied: legacy-api-wrap>=1.4.1 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (1.5)
Requirement already satisfied: matplotlib>=3.7.5 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
(3.10.0)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from scanpy>=1.7.0->celltypist) (8.4.0)
Requirement already satisfied: networkx>=2.7.1 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (3.5)
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Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
(0.60.0)
Requirement already satisfied: packaging>=21.3 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (25.0)
Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-
packages (from scanpy>=1.7.0->celltypist) (1.0.2)
Requirement already satisfied: pynndescent>=0.5.13 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
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Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-
packages (from scanpy>=1.7.0->celltypist) (1.16.3)
Requirement already satisfied: seaborn>=0.13.2 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
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Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: statsmodels>=0.14.5 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
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Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
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Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
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Requirement already satisfied: umap-learn>=0.5.6 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
(0.5.9.post2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.1->celltypist)
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Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from
anndata>=0.8->scanpy>=1.7.0->celltypist) (1.12.0)
Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
/usr/local/lib/python3.12/dist-packages (from
anndata>=0.8->scanpy>=1.7.0->celltypist) (3.1.5)
Requirement already satisfied: texttable>=1.6.2 in
/usr/local/lib/python3.12/dist-packages (from
igraph<2.0,>=1.0.0->leidenalg>=0.9.0->celltypist) (1.7.0)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (4.60.1)
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Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (3.2.5)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from
numba!=0.62.0rc1,>=0.57.1->scanpy>=1.7.0->celltypist) (0.43.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->pandas>=1.0.5->celltypist) (1.17.0)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-
packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist)
(0.8.1.post1)
Requirement already satisfied: google-crc32c>=1.5 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (1.7.1)
Requirement already satisfied: numcodecs>=0.14 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.16.5)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from
donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist)
(6.0.3)
Downloading celltypist-1.7.1-py3-none-any.whl (7.3 MB)
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107.8 MB/s eta 0:00:00
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leidenalg-0.11.0-cp38-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (2.7
MB)
                         2.7/2.7 MB
83.6 MB/s eta 0:00:00
Installing collected packages: leidenalg, celltypist
Successfully installed celltypist-1.7.1 leidenalg-0.11.0
Collecting decoupler
  Downloading decoupler-2.1.2-py3-none-any.whl.metadata (8.8 kB)
Collecting adjusttext (from decoupler)
  Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
packages (from decoupler) (0.12.6)
Collecting docrep (from decoupler)
  Downloading docrep-0.3.2.tar.gz (33 kB)
  Preparing metadata (setup.py) … done
Collecting marsilea (from decoupler)
  Downloading marsilea-0.5.6-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages
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(from decoupler) (0.60.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
packages (from decoupler) (2.32.4)
Collecting scipy<1.16 (from decoupler)
  Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
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Requirement already satisfied: session-info2 in
/usr/local/lib/python3.12/dist-packages (from decoupler) (0.2.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from decoupler) (4.67.1)
Requirement already satisfied: numpy<2.5,>=1.23.5 in
/usr/local/lib/python3.12/dist-packages (from scipy<1.16->decoupler) (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from adjusttext->decoupler) (3.10.0)
Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.12.0)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (3.15.1)
Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.5)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (8.4.0)
Requirement already satisfied: packaging>=24.2 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (25.0)
Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (2.2.2)
Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.1.5)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from docrep->decoupler) (1.17.0)
Collecting legendkit (from marsilea->decoupler)
  Downloading legendkit-0.3.6-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (4.5.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (0.13.2)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba->decoupler) (0.43.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->decoupler) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
```

```
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2025.11.12)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->adjusttext->decoupler) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->adjusttext->decoupler) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-
packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.8.1.post1)
Requirement already satisfied: google-crc32c>=1.5 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (1.7.1)
Requirement already satisfied: numcodecs>=0.14 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.16.5)
Requirement already satisfied: typing-extensions>=4.9 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (4.15.0)
Requirement already satisfied: pyarrow>=10.0.1 in
/usr/local/lib/python3.12/dist-packages (from
pandas[parquet]->marsilea->decoupler) (18.1.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)
Downloading decoupler-2.1.2-py3-none-any.whl (120 kB)
                         120.9/120.9 kB
10.7 MB/s eta 0:00:00
Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3
MB)
                         37.3/37.3 MB
24.2 MB/s eta 0:00:00
```

```
Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)
Downloading marsilea-0.5.6-py3-none-any.whl (84 kB)
                         85.0/85.0 kB
6.1 MB/s eta 0:00:00
Downloading legendkit-0.3.6-py3-none-any.whl (24 kB)
Building wheels for collected packages: docrep
  Building wheel for docrep (setup.py) … done
  Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19876
sha256=ad25052de0d26b42a2b305de25b8bfc6ec6c866c5137c04948a4d86ce12a1c42
  Stored in directory: /root/.cache/pip/wheels/d6/19/ee/0a6a1793d91c449563b49cca
b57ce52da3e6fab7614836bd8c
Successfully built docrep
Installing collected packages: scipy, docrep, legendkit, adjusttext, marsilea,
decoupler
  Attempting uninstall: scipy
    Found existing installation: scipy 1.16.3
    Uninstalling scipy-1.16.3:
      Successfully uninstalled scipy-1.16.3
Successfully installed adjusttext-1.3.0 decoupler-2.1.2 docrep-0.3.2
legendkit-0.3.6 marsilea-0.5.6 scipy-1.15.3
Collecting fa2-modified
  Downloading fa2_modified-0.4-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (9.3
kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (1.15.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (4.67.1)
Downloading fa2_modified-0.4-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (620 kB)
                         620.8/620.8 kB
28.5 MB/s eta 0:00:00
Installing collected packages: fa2-modified
Successfully installed fa2-modified-0.4
Collecting louvain
  Downloading louvain-0.8.2.tar.gz (4.2 MB)
                           4.2/4.2 MB
73.3 MB/s eta 0:00:00
  Preparing metadata (setup.py) … done
Collecting igraph<0.12,>=0.10.0 (from louvain)
  Downloading
igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(4.4 kB)
Requirement already satisfied: texttable>=1.6.2 in
/usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->louvain)
(1.7.0)
```

```
Downloading
igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.4 MB)
                         4.4/4.4 MB
80.7 MB/s eta 0:00:00
Building wheels for collected packages: louvain
  Building wheel for louvain (setup.py) … done
  Created wheel for louvain: filename=louvain-0.8.2-cp312-cp312-linux_x86_64.whl
size=971849
sha256=841ff463361477ec9f55149d272f4f2b0e19ff2162cfee1cbd1e709ec0de641a
  Stored in directory: /root/.cache/pip/wheels/40/de/2b/bb7ed19d84727f9f299f20cd
34c42bba9c8bef7d83d2255c86
Successfully built louvain
Installing collected packages: igraph, louvain
  Attempting uninstall: igraph
    Found existing installation: igraph 1.0.0
    Uninstalling igraph-1.0.0:
      Successfully uninstalled igraph-1.0.0
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the following
dependency conflicts.
leidenalg 0.11.0 requires igraph<2.0,>=1.0.0, but you have igraph 0.11.9 which
is incompatible.
Successfully installed igraph-0.11.9 louvain-0.8.2
```

[ ]:

```
Collecting decoupler
  Downloading decoupler-2.1.2-py3-none-any.whl.metadata (8.8 kB)
Collecting adjusttext (from decoupler)
  Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)
Collecting anndata (from decoupler)
  Downloading anndata-0.12.6-py3-none-any.whl.metadata (10.0 kB)
Collecting docrep (from decoupler)
  Downloading docrep-0.3.2.tar.gz (33 kB)
  Preparing metadata (setup.py) … done
Collecting marsilea (from decoupler)
  Downloading marsilea-0.5.6-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages
(from decoupler) (0.60.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
packages (from decoupler) (2.32.4)
Collecting scipy<1.16 (from decoupler)
  Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(61 kB)
```

```
                              62.0/62.0 kB
4.3 MB/s eta 0:00:00
Collecting session-info2 (from decoupler)
  Downloading session_info2-0.2.3-py3-none-any.whl.metadata (3.4 kB)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from decoupler) (4.67.1)
Requirement already satisfied: numpy<2.5,>=1.23.5 in
/usr/local/lib/python3.12/dist-packages (from scipy<1.16->decoupler) (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from adjusttext->decoupler) (3.10.0)
Collecting array-api-compat>=1.7.1 (from anndata->decoupler)
  Downloading array_api_compat-1.12.0-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (3.15.1)
Collecting legacy-api-wrap (from anndata->decoupler)
  Downloading legacy_api_wrap-1.5-py3-none-any.whl.metadata (2.2 kB)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (8.4.0)
Requirement already satisfied: packaging>=24.2 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (25.0)
Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (2.2.2)
Collecting zarr!=3.0.*,>=2.18.7 (from anndata->decoupler)
  Downloading zarr-3.1.5-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from docrep->decoupler) (1.17.0)
Collecting legendkit (from marsilea->decoupler)
  Downloading legendkit-0.3.6-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (4.5.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (0.13.2)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba->decoupler) (0.43.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->decoupler) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2025.11.12)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->adjusttext->decoupler) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
```

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/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->adjusttext->decoupler) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)
Collecting donfig>=0.8 (from zarr!=3.0.*,>=2.18.7->anndata->decoupler)
  Downloading donfig-0.8.1.post1-py3-none-any.whl.metadata (5.0 kB)
Requirement already satisfied: google-crc32c>=1.5 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (1.7.1)
Collecting numcodecs>=0.14 (from zarr!=3.0.*,>=2.18.7->anndata->decoupler)
  Downloading numcodecs-0.16.5-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata
(3.4 kB)
Requirement already satisfied: typing-extensions>=4.9 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (4.15.0)
Requirement already satisfied: pyarrow>=10.0.1 in
/usr/local/lib/python3.12/dist-packages (from
pandas[parquet]->marsilea->decoupler) (18.1.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)
Downloading decoupler-2.1.2-py3-none-any.whl (120 kB)
                         120.9/120.9 kB
8.4 MB/s eta 0:00:00
Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3
MB)
                         37.3/37.3 MB
52.1 MB/s eta 0:00:00
Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)
Downloading anndata-0.12.6-py3-none-any.whl (172 kB)
                         172.3/172.3 kB
12.9 MB/s eta 0:00:00
Downloading marsilea-0.5.6-py3-none-any.whl (84 kB)
                         85.0/85.0 kB
```

```
5.7 MB/s eta 0:00:00
Downloading session_info2-0.2.3-py3-none-any.whl (16 kB)
Downloading array_api_compat-1.12.0-py3-none-any.whl (58 kB)
                         58.2/58.2 kB
3.5 MB/s eta 0:00:00
Downloading zarr-3.1.5-py3-none-any.whl (284 kB)
                         284.1/284.1 kB
18.6 MB/s eta 0:00:00
Downloading legacy_api_wrap-1.5-py3-none-any.whl (10 kB)
Downloading legendkit-0.3.6-py3-none-any.whl (24 kB)
Downloading donfig-0.8.1.post1-py3-none-any.whl (21 kB)
Downloading numcodecs-0.16.5-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (9.2 MB)
                         9.2/9.2 MB
75.3 MB/s eta 0:00:00
Building wheels for collected packages: docrep
  Building wheel for docrep (setup.py) … done
  Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19876
sha256=3c7a0b9d48d1508a4d1090c7a199af14059a0307238a411f338d20b37f188ca3
  Stored in directory: /root/.cache/pip/wheels/d6/19/ee/0a6a1793d91c449563b49cca
b57ce52da3e6fab7614836bd8c
Successfully built docrep
Installing collected packages: session-info2, scipy, numcodecs, legacy-api-wrap,
donfig, docrep, array-api-compat, zarr, legendkit, anndata, adjusttext,
marsilea, decoupler
  Attempting uninstall: scipy
    Found existing installation: scipy 1.16.3
    Uninstalling scipy-1.16.3:
      Successfully uninstalled scipy-1.16.3
Successfully installed adjusttext-1.3.0 anndata-0.12.6 array-api-compat-1.12.0
decoupler-2.1.2 docrep-0.3.2 donfig-0.8.1.post1 legacy-api-wrap-1.5
legendkit-0.3.6 marsilea-0.5.6 numcodecs-0.16.5 scipy-1.15.3 session-info2-0.2.3
zarr-3.1.5
```

[ ]:
```python
#Import core single cell datasets

import scanpy as sc
import anndata as ad
import numpy as np
```

[ ]:
```
!wget https://ftp.ncbi.nlm.nih.gov/geo/series/GSE166nnn/GSE166766/suppl/
↪GSE166766%5Fscv2%5F200428.h5ad.gz
```

```
--2025-11-24 11:28:22--  https://ftp.ncbi.nlm.nih.gov/geo/series/GSE166nnn/GSE16
6766/suppl/GSE166766%5Fscv2%5F200428.h5ad.gz
Resolving ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)… 130.14.250.31,
130.14.250.10, 130.14.250.11, …
Connecting to ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)|130.14.250.31|:443…
```

```
connected.
HTTP request sent, awaiting response… 200 OK
Length: 13440995733 (13G) [application/x-gzip]
Saving to: ‘GSE166766_scv2_200428.h5ad.gz’

GSE166766_scv2_2004 100%[===================>]  12.52G  17.0MB/s    in 13m 17s

2025-11-24 11:41:40 (16.1 MB/s) - ‘GSE166766_scv2_200428.h5ad.gz’ saved
[13440995733/13440995733]
```

[ ]:
```python
bm_mtx = sc.read_h5ad('/content/497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad')
bm_mtx.var_names_make_unique()
```

[ ]:
```python
bm_mtx = sc.read_h5ad('/content/GSE166766_scv2_200428.h5ad.gz')
bm_mtx.var_names_make_unique()
```

```
---------------------------------------------------------------------------
OSError                                   Traceback (most recent call last)
/tmp/ipython-input-2626531853.py in <cell line: 0>()
----> 1 bm_mtx = sc.read_h5ad('/content/GSE166766_scv2_200428.h5ad.gz')
      2 bm_mtx.var_names_make_unique()

/usr/local/lib/python3.12/dist-packages/anndata/_io/h5ad.py in␣
↪read_h5ad(filename, backed, as_sparse, as_sparse_fmt, chunk_size)
    260     )
    261
--> 262     with h5py.File(filename, "r") as f:
    263
    264         def callback(func, elem_name: str, elem, iospec):

/usr/local/lib/python3.12/dist-packages/h5py/_hl/files.py in __init__(self,␣
↪name, mode, driver, libver, userblock_size, swmr, rdcc_nslots, rdcc_nbytes,␣
↪rdcc_w0, track_order, fs_strategy, fs_persist, fs_threshold, fs_page_size,␣
↪page_buf_size, min_meta_keep, min_raw_keep, locking, alignment_threshold,␣
↪alignment_interval, meta_block_size, track_times, **kwds)
    564                                  fs_strategy=fs_strategy,␣
↪fs_persist=fs_persist,
    565                                  fs_threshold=fs_threshold,␣
↪fs_page_size=fs_page_size)
--> 566                 fid = make_fid(name, mode, userblock_size, fapl, fcpl,␣
↪swmr=swmr)
    567
    568             if isinstance(libver, tuple):

/usr/local/lib/python3.12/dist-packages/h5py/_hl/files.py in make_fid(name,␣
↪mode, userblock_size, fapl, fcpl, swmr)
    239         if swmr and swmr_support:
```

```
    240             flags |= h5f.ACC_SWMR_READ
--> 241         fid = h5f.open(name, flags, fapl=fapl)
    242     elif mode == 'r+':
    243         fid = h5f.open(name, h5f.ACC_RDWR, fapl=fapl)

h5py/_objects.pyx in h5py._objects.with_phil.wrapper()

h5py/_objects.pyx in h5py._objects.with_phil.wrapper()

h5py/h5f.pyx in h5py.h5f.open()

OSError: Unable to synchronously open file (file signature not found)
```

[ ]:

[ ]:

[ ]:

[ ]:
```
#!wget https://datasets.cellxgene.cziscience.com/
↪fdf57c52-ad71-4004-9db2-a962e849b524.h5ad
!wget https://datasets.cellxgene.cziscience.com/
↪497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad
```

```
--2025-11-20 14:25:29--  https://datasets.cellxgene.cziscience.com/497ab773-
4fd5-4263-9fdf-7b42a68f1351.h5ad
Resolving datasets.cellxgene.cziscience.com
(datasets.cellxgene.cziscience.com)… 3.165.160.82, 3.165.160.128,
3.165.160.58, …
Connecting to datasets.cellxgene.cziscience.com
(datasets.cellxgene.cziscience.com)|3.165.160.82|:443… connected.
HTTP request sent, awaiting response… 200 OK
Length: 74549927 (71M) [binary/octet-stream]
Saving to: ‘497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad’

497ab773-4fd5-4263- 100%[===================>]  71.10M  63.3MB/s    in 1.1s

2025-11-20 14:25:30 (63.3 MB/s) - ‘497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad’
saved [74549927/74549927]
```

[ ]:
```
"""
mkdir -p GSM6229474
mkdir -p GSM6229475
mv GSM6229474_* GSM6229474
mv GSM6229475_* GSM6229475
cd GSM6229474
```

```
mv GSM6229474_C57IMQD_barcodes.tsv.gz barcodes.tsv.gz
mv GSM6229474_C57IMQD_features.tsv.gz features.tsv.gz
mv GSM6229474_C57IMQD_matrix.mtx.gz matrix.mtx.gz
cd ..
cd GSM6229475
mv GSM6229475_C57IMQE_barcodes.tsv.gz barcodes.tsv.gz
mv GSM6229475_C57IMQE_features.tsv.gz features.tsv.gz
mv GSM6229475_C57IMQE_matrix.mtx.gz matrix.mtx.gz
cd ..
"""
```

[ ]:
```
'\nmkdir -p GSM6229474\nmkdir -p GSM6229475\nmv GSM6229474_* GSM6229474\nmv
GSM6229475_* GSM6229475\ncd GSM6229474\nmv GSM6229474_C57IMQD_barcodes.tsv.gz
barcodes.tsv.gz\nmv GSM6229474_C57IMQD_features.tsv.gz features.tsv.gz\nmv
GSM6229474_C57IMQD_matrix.mtx.gz matrix.mtx.gz\ncd ..\ncd GSM6229475\nmv
GSM6229475_C57IMQE_barcodes.tsv.gz barcodes.tsv.gz\nmv
GSM6229475_C57IMQE_features.tsv.gz features.tsv.gz\nmv
GSM6229475_C57IMQE_matrix.mtx.gz matrix.mtx.gz\ncd ..\n'
```

[ ]:
```
bm_mtx = sc.read_h5ad('/content/497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad')
bm_mtx.var_names_make_unique()
```

[ ]:
```
bm_mtx
```

[ ]:
```
AnnData object with n_obs × n_vars = 14783 × 17374
    obs: 'disease stage', 'treatment', 'timepoint', 'Dataset', 'sample',
'disease_original', 'disease_general', 'COVID-19 Condition', 'Lineage',
'Cell.group', 'Cell.class_reannotated', 'n_genes', 'n_counts', 'percent_mito',
'tissue_original', 'tissue_ontology_term_id', 'disease_ontology_term_id',
'donor_id', 'development_stage_ontology_term_id', 'assay_ontology_term_id',
'cell_type_ontology_term_id', 'self_reported_ethnicity_ontology_term_id',
'sex_ontology_term_id', 'is_primary_data', 'suspension_type', 'tissue_type',
'cell_type', 'assay', 'disease', 'sex', 'tissue', 'self_reported_ethnicity',
'development_stage', 'observation_joinid'
    var: 'n_cells', 'feature_is_filtered', 'feature_name', 'feature_reference',
'feature_biotype', 'feature_length', 'feature_type'
    uns: 'citation', 'doi', 'organism', 'organism_ontology_term_id',
'schema_reference', 'schema_version', 'title'
    obsm: 'X_pca', 'X_tsne', 'X_umap'
```

[ ]:
```
del bm_mtx.obs['cell_type']
```

[ ]:
```
bm_mtx
```

[ ]:
```
AnnData object with n_obs × n_vars = 14783 × 17374
    obs: 'disease stage', 'treatment', 'timepoint', 'Dataset', 'sample',
'disease_original', 'disease_general', 'COVID-19 Condition', 'Lineage',
```

```
'Cell.group', 'Cell.class_reannotated', 'n_genes', 'n_counts', 'percent_mito',
'tissue_original', 'tissue_ontology_term_id', 'disease_ontology_term_id',
'donor_id', 'development_stage_ontology_term_id', 'assay_ontology_term_id',
'cell_type_ontology_term_id', 'self_reported_ethnicity_ontology_term_id',
'sex_ontology_term_id', 'is_primary_data', 'suspension_type', 'tissue_type',
'assay', 'disease', 'sex', 'tissue', 'self_reported_ethnicity',
'development_stage', 'observation_joinid'
    var: 'n_cells', 'feature_is_filtered', 'feature_name', 'feature_reference',
'feature_biotype', 'feature_length', 'feature_type'
    uns: 'citation', 'doi', 'organism', 'organism_ontology_term_id',
'schema_reference', 'schema_version', 'title'
    obsm: 'X_pca', 'X_tsne', 'X_umap'
```

[ ]:
```
bm_mtx.write_h5ad("bone_marrow.h5ad")
```

[ ]:
```
short_read = sc.read_h5ad('bone_marrow.h5ad')
```

[ ]:
```
short_read
```

[ ]:
```
AnnData object with n_obs × n_vars = 14783 × 17374
    obs: 'disease stage', 'treatment', 'timepoint', 'Dataset', 'sample',
'disease_original', 'disease_general', 'COVID-19 Condition', 'Lineage',
'Cell.group', 'Cell.class_reannotated', 'n_genes', 'n_counts', 'percent_mito',
'tissue_original', 'tissue_ontology_term_id', 'disease_ontology_term_id',
'donor_id', 'development_stage_ontology_term_id', 'assay_ontology_term_id',
'cell_type_ontology_term_id', 'self_reported_ethnicity_ontology_term_id',
'sex_ontology_term_id', 'is_primary_data', 'suspension_type', 'tissue_type',
'assay', 'disease', 'sex', 'tissue', 'self_reported_ethnicity',
'development_stage', 'observation_joinid'
    var: 'n_cells', 'feature_is_filtered', 'feature_name', 'feature_reference',
'feature_biotype', 'feature_length', 'feature_type'
    uns: 'citation', 'doi', 'organism', 'organism_ontology_term_id',
'schema_reference', 'schema_version', 'title'
    obsm: 'X_pca', 'X_tsne', 'X_umap'
```

[ ]:
```
bm_mtx.shape
```

[ ]:
```
(14783, 17374)
```

[ ]:
```
# A useful step for older datasets
bm_mtx.var_names_make_unique()
bm_mtx.obs_names_make_unique()
```

[ ]:
```
bm_mtx.var.head()
```

[ ]:
```
                 n_cells  feature_is_filtered feature_name feature_reference  \
ENSG00000161920      927                False        MED11    NCBITaxon:9606
```

```
ENSG00000122335      429               False        SERAC1     NCBITaxon:9606
ENSG00000175548      149               False        ALG10B     NCBITaxon:9606
ENSG00000100330      426               False         MTMR3     NCBITaxon:9606
ENSG00000176340     7171               False         COX8A     NCBITaxon:9606

                feature_biotype feature_length    feature_type
ENSG00000161920            gene            754  protein_coding
ENSG00000122335            gene           2683  protein_coding
ENSG00000175548            gene           1773  protein_coding
ENSG00000100330            gene            634  protein_coding
ENSG00000176340            gene            494  protein_coding
```

[ ]:
```
bm_mtx.shape
```

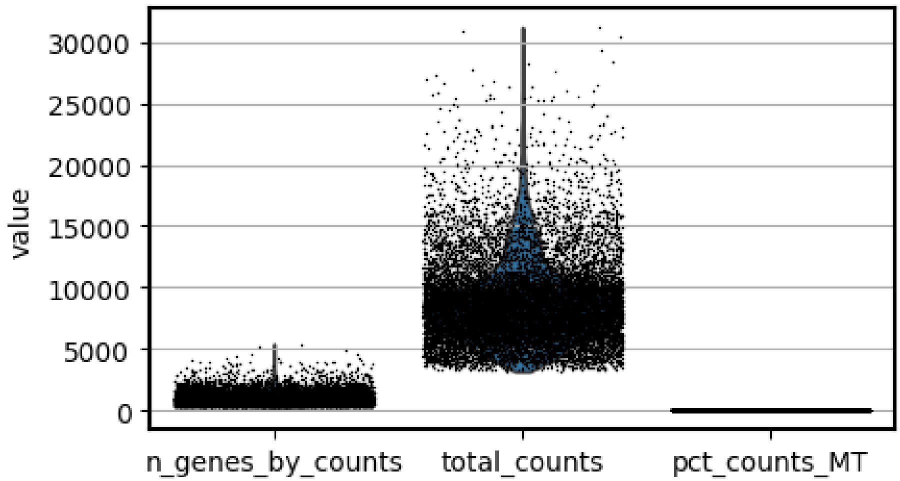
[ ]: (14783, 17374)

[ ]:
```
bm_mtx.var['MT'] = bm_mtx.var_names.str.startswith("MT-")
bm_mtx.var['RIBO'] = bm_mtx.var_names.str.startswith("RPS", "RPL")
bm_mtx.var['HB'] = bm_mtx.var_names.str.startswith("^HB[^(P)]")
```

[ ]:
```
sc.pp.calculate_qc_metrics(
    bm_mtx, qc_vars=["MT", 'RIBO', 'HB'], inplace=True, log1p=True
)
```

[ ]:
```
import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (5,3)  # Adjust figure size
plt.rcParams["axes.grid"] = True  # Add grid to plots
plt.rcParams["axes.edgecolor"] = "black" # Set plot border color
plt.rcParams["axes.linewidth"] = 1.5 # Set plot border width
plt.rcParams["axes.facecolor"] = "white" # Set background color
plt.rcParams["axes.labelcolor"] = "black" # Set label color
plt.rcParams["xtick.color"] = "black" # Set x-axis tick color
plt.rcParams["ytick.color"] = "black" # Set y-axis tick color
plt.rcParams["text.color"] = "black" # Set text color
```

[ ]:
```
sc.pl.violin(
    bm_mtx,
    ["n_genes_by_counts", 'total_counts', 'pct_counts_MT'],
    jitter=0.4,
    multi_panel=False,
)
```



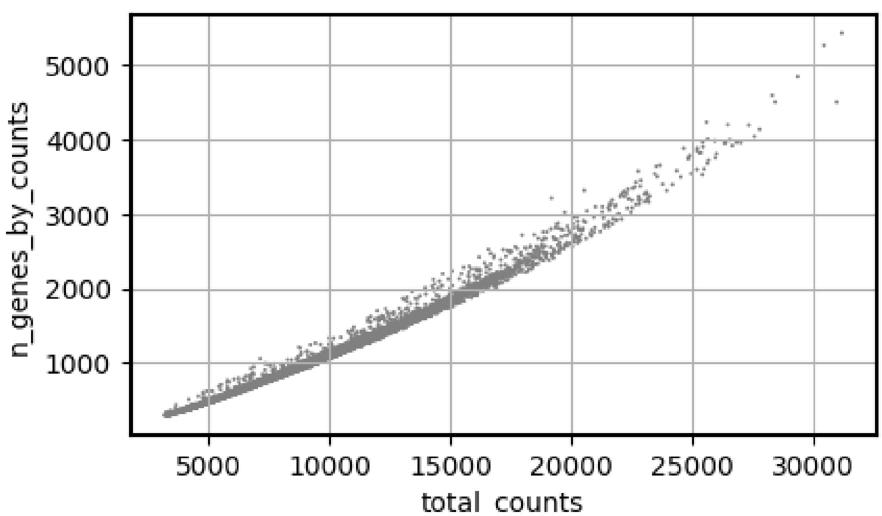
[ ]:
```
bm_mtx.obs_keys()
```

```
/tmp/ipython-input-3505303993.py:1: FutureWarning: Use obs (e.g. `k in
adata.obs` or `str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys,
AnnData.obs_keys is deprecated and will be removed in the future.
  bm_mtx.obs_keys()
```

[ ]:
```
['disease stage',
 'treatment',
 'timepoint',
 'Dataset',
 'sample',
 'disease_original',
 'disease_general',
 'COVID-19 Condition',
 'Lineage',
 'Cell.group',
 'Cell.class_reannotated',
 'n_genes',
 'n_counts',
 'percent_mito',
 'tissue_original',
 'tissue_ontology_term_id',
 'disease_ontology_term_id',
 'donor_id',
 'development_stage_ontology_term_id',
 'assay_ontology_term_id',
 'cell_type_ontology_term_id',
 'self_reported_ethnicity_ontology_term_id',
 'sex_ontology_term_id',
```

```
 'is_primary_data',
 'suspension_type',
 'tissue_type',
 'assay',
 'disease',
 'sex',
 'tissue',
 'self_reported_ethnicity',
 'development_stage',
 'observation_joinid',
 'n_genes_by_counts',
 'log1p_n_genes_by_counts',
 'total_counts',
 'log1p_total_counts',
 'pct_counts_in_top_50_genes',
 'pct_counts_in_top_100_genes',
 'pct_counts_in_top_200_genes',
 'pct_counts_in_top_500_genes',
 'total_counts_MT',
 'log1p_total_counts_MT',
 'pct_counts_MT',
 'total_counts_RIBO',
 'log1p_total_counts_RIBO',
 'pct_counts_RIBO',
 'total_counts_HB',
 'log1p_total_counts_HB',
 'pct_counts_HB']
```

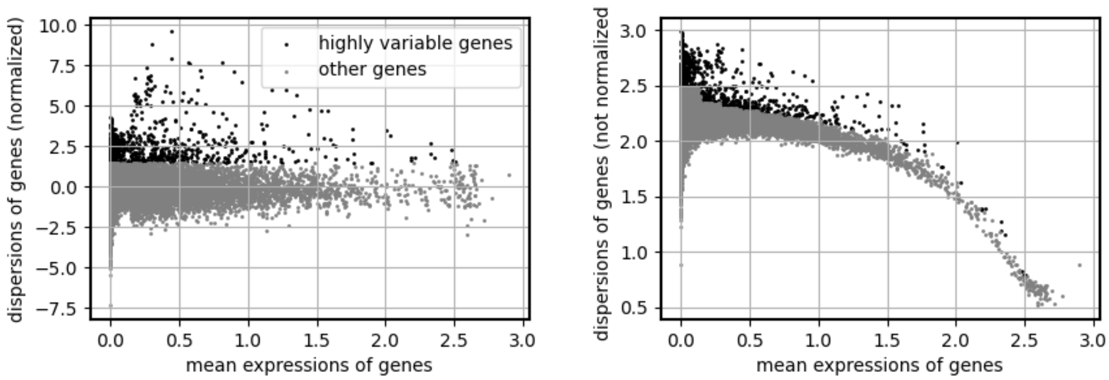
[ ]:
```
sc.pl.scatter(bm_mtx, "total_counts", "n_genes_by_counts")#, color="cell_type")
```



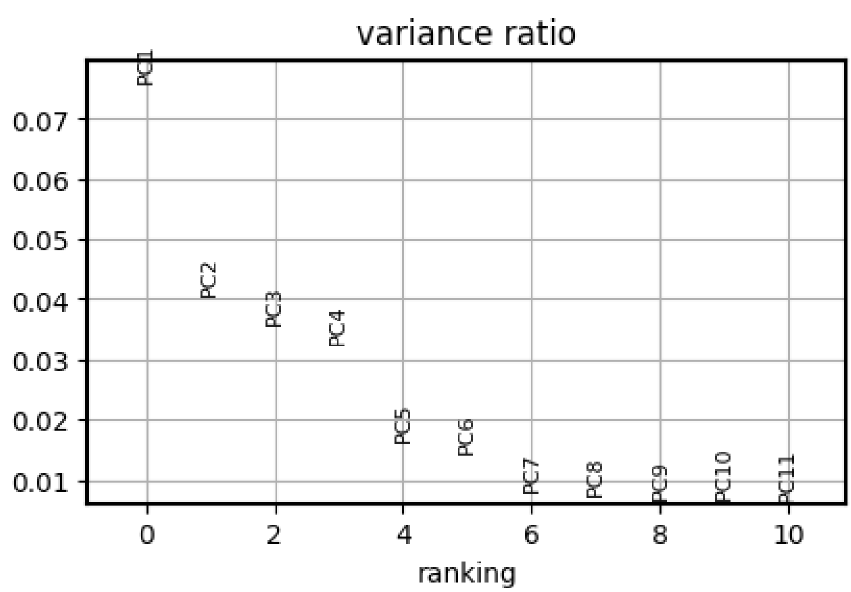
```
[ ]: #sc.pp.scrublet(bm_mtx)
```

```
[ ]: #Normalisation
     bm_mtx.layers["counts"] = bm_mtx.X.copy()
     sc.pp.normalize_total(bm_mtx)
     sc.pp.log1p(bm_mtx)
```

```
[ ]: #Feature selection
     sc.pp.highly_variable_genes(bm_mtx, n_top_genes=1000)
     sc.pl.highly_variable_genes(bm_mtx)
```




```
[ ]: #Dim Reduction
     sc.tl.pca(bm_mtx)
     sc.pl.pca_variance_ratio(bm_mtx, n_pcs=10, log=False)
```



```
[ ]: bm_mtx.var_names
```

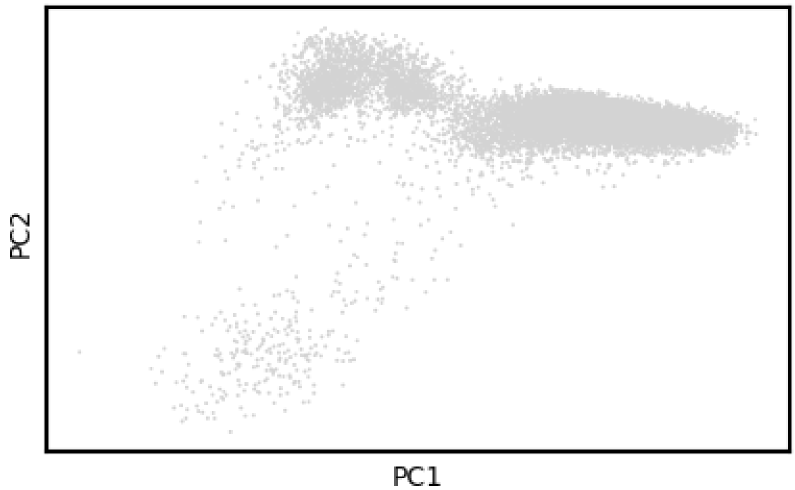
```
[ ]: Index(['ENSG00000161920', 'ENSG00000122335', 'ENSG00000175548',
            'ENSG00000100330', 'ENSG00000176340', 'ENSG00000179846',
            'ENSG00000204860', 'ENSG00000172260', 'ENSG00000141424',
            'ENSG00000164512',
            …
            'ENSG00000164114', 'ENSG00000151702', 'ENSG00000224578',
            'ENSG00000138756', 'ENSG00000111052', 'ENSG00000176946',
            'ENSG00000150456', 'ENSG00000284934', 'ENSG00000261842',
            'ENSG00000260456'],
           dtype='object', length=17374)
```

```
[ ]: #il_genes = [gene for gene in bm_mtx.var_names[bm_mtx.var['highly_variable']]
      ↪if gene.lower().startswith('cc')]
     #print(il_genes)
```

```
[ ]: bm_mtx
```

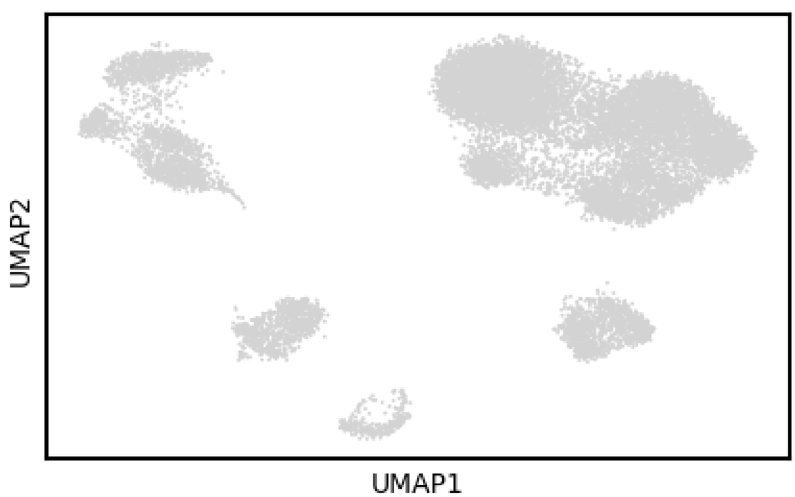
```
[ ]: AnnData object with n_obs × n_vars = 14783 × 17374
         obs: 'disease stage', 'treatment', 'timepoint', 'Dataset', 'sample',
     'disease_original', 'disease_general', 'COVID-19 Condition', 'Lineage',
     'Cell.group', 'Cell.class_reannotated', 'n_genes', 'n_counts', 'percent_mito',
     'tissue_original', 'tissue_ontology_term_id', 'disease_ontology_term_id',
     'donor_id', 'development_stage_ontology_term_id', 'assay_ontology_term_id',
     'cell_type_ontology_term_id', 'self_reported_ethnicity_ontology_term_id',
     'sex_ontology_term_id', 'is_primary_data', 'suspension_type', 'tissue_type',
     'assay', 'disease', 'sex', 'tissue', 'self_reported_ethnicity',
     'development_stage', 'observation_joinid', 'n_genes_by_counts',
     'log1p_n_genes_by_counts', 'total_counts', 'log1p_total_counts',
     'pct_counts_in_top_50_genes', 'pct_counts_in_top_100_genes',
     'pct_counts_in_top_200_genes', 'pct_counts_in_top_500_genes', 'total_counts_MT',
     'log1p_total_counts_MT', 'pct_counts_MT', 'total_counts_RIBO',
     'log1p_total_counts_RIBO', 'pct_counts_RIBO', 'total_counts_HB',
     'log1p_total_counts_HB', 'pct_counts_HB'
         var: 'n_cells', 'feature_is_filtered', 'feature_name', 'feature_reference',
     'feature_biotype', 'feature_length', 'feature_type', 'MT', 'RIBO', 'HB',
     'n_cells_by_counts', 'mean_counts', 'log1p_mean_counts',
     'pct_dropout_by_counts', 'total_counts', 'log1p_total_counts',
     'highly_variable', 'means', 'dispersions', 'dispersions_norm'
         uns: 'citation', 'doi', 'organism', 'organism_ontology_term_id',
     'schema_reference', 'schema_version', 'title', 'log1p', 'hvg', 'pca'
         obsm: 'X_pca', 'X_tsne', 'X_umap'
         varm: 'PCs'
         layers: 'counts'
```

```
[ ]: sc.pl.pca(bm_mtx,
               #color=["cell_type"],
               cmap="coolwarm")
```



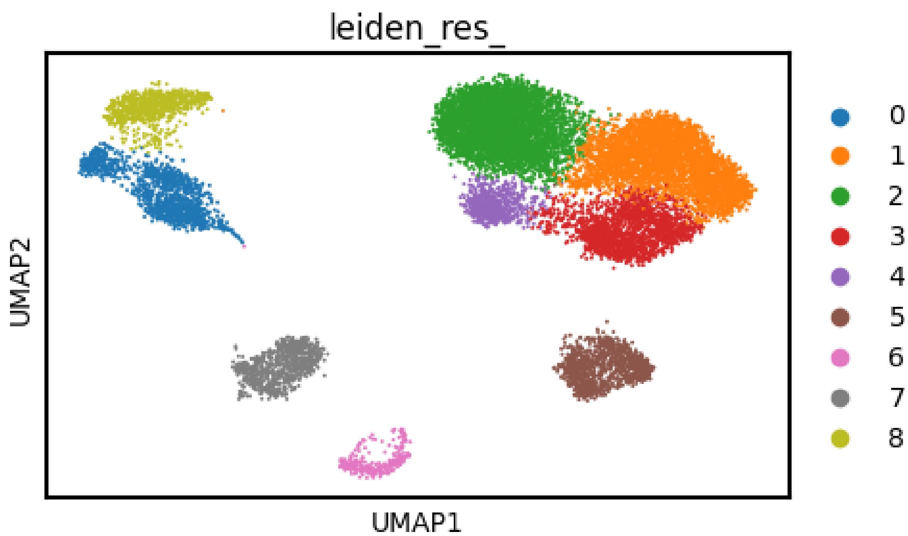

```
[ ]: sc.pp.neighbors(bm_mtx)
     sc.tl.umap(bm_mtx)
```

```
[ ]: sc.pl.umap(
         bm_mtx,
        #color=["cell_type"],
         size=8,
     )
```



[ ]:
```python
sc.tl.leiden(bm_mtx, flavor="igraph", n_iterations=10, key_added="leiden_res_", resolution=0.5 )
```

[ ]:
```python
sc.pl.umap(
    bm_mtx,
    color=["leiden_res_"],
    size=8,
)
```



[ ]:
```python
import decoupler as dc
```

[ ]:
```python
!wget wget -O result.txt 'http://www.ensembl.org/biomart/martservice?query=<?xml version="1.0" encoding="UTF-8"?><!DOCTYPE Query><Query  virtualSchemaName = "default" formatter = "CSV" header = "0" uniqueRows = "0" count = "" datasetConfigVersion = "0.6" ><Dataset name = "hsapiens_gene_ensembl" interface = "default" ><Attribute name = "ensembl_gene_id" /><Attribute name = "external_gene_name" /></Dataset></Query>'
```

```
--2025-11-14 22:01:29--  http://wget/
Resolving wget (wget)… failed: Name or service not known.
wget: unable to resolve host address ‘wget’
--2025-11-14 22:01:30--  http://www.ensembl.org/biomart/martservice?query=%3C?xm
l%20version=%221.0%22%20encoding=%22UTF-
8%22?%3E%3C!DOCTYPE%20Query%3E%3CQuery%20%20virtualSchemaName%20=%20%22default%2
2%20formatter%20=%20%22CSV%22%20header%20=%20%220%22%20uniqueRows%20=%20%220%22%
20count%20=%20%22%22%20datasetConfigVersion%20=%20%220.6%22%20%3E%3CDataset%20na
me%20=%20%22hsapiens_gene_ensembl%22%20interface%20=%20%22default%22%20%3E%3CAtt
ribute%20name%20=%20%22ensembl_gene_id%22%20/%3E%3CAttribute%20name%20=%20%22ext
```

```
ernal_gene_name%22%20/%3E%3C/Dataset%3E%3C/Query%3E
Resolving www.ensembl.org (www.ensembl.org)… 193.62.193.83
Connecting to www.ensembl.org (www.ensembl.org)|193.62.193.83|:80… connected.
HTTP request sent, awaiting response… 200 OK
Length: unspecified [text/plain]
Saving to: 'result.txt'

result.txt              [       <=>          ]   1.71M  1.39MB/s    in 1.2s

2025-11-14 22:01:31 (1.39 MB/s) - 'result.txt' saved [1795328]

FINISHED --2025-11-14 22:01:31--
Total wall clock time: 1.8s
Downloaded: 1 files, 1.7M in 1.2s (1.39 MB/s)
```

[ ]:
```
import pandas as pd

ensembl_var = pd.read_csv('/content/result.txt', header = None)

ensembl_var.columns = ['ensembl_gene_id', 'gene_name']

ensembl_var.head(3)
```

[ ]:
```
   ensembl_gene_id gene_name
0  ENSG00000210049     MT-TF
1  ENSG00000211459   MT-RNR1
2  ENSG00000210077     MT-TV
```

[ ]:
```
markers = dc.op.resource(name="PanglaoDB", organism="human")
```

[ ]:
```
markers.head()
```

[ ]:
```
  genesymbol  canonical_marker      cell_type germ_layer  human  \

0       A1CF             False    Hepatocytes   Endoderm   True
1     A2APA5             False     Germ cells   Mesoderm  False
2        A2M              True  Bergmann glia   Ectoderm   True
3     A3FIN4             False     Mast cells   Mesoderm  False
4     A4GALT              True     Mast cells   Mesoderm   True

   human_sensitivity  human_specificity  mouse  mouse_sensitivity  \

0           0.189189           0.004437   True           0.175000
1           0.000000           0.000000   True           0.432927
2           0.000000           0.062343   True           0.333333
3           0.000000           0.000000   True           0.184211
4           0.000000           0.014133   True           0.000000
```

	mouse_specificity	ncbi_tax_id	organ	ubiquitiousness
0	0.000257	9606	Liver	0.002
1	0.000000	10090	Reproductive	0.004
2	0.001604	9606	Brain	0.012
3	0.000708	10090	Immune system	0.001
4	0.003023	9606	Immune system	0.005

[ ]:
```
markers.organ.unique()
```

[ ]:
```
array(['Liver', 'Reproductive', 'Brain', 'Immune system', 'Zygote',
       'Kidney', 'Blood', 'Lungs', 'Bone', 'GI tract', nan, 'Vasculature',
       'Pancreas', 'Heart', 'Mammary gland', 'Olfactory system',
       'Connective tissue', 'Epithelium', 'Skeletal muscle', 'Skin',
       'Embryo', 'Smooth muscle', 'Eye', 'Adrenal glands', 'Thyroid',
       'Placenta', 'Thymus', 'Parathyroid glands', 'Oral cavity',
       'Urinary bladder'], dtype=object)
```

[ ]:
```

```

[ ]:
```
# Query Omnipath and get PanglaoDB
markers = dc.op.resource(name="PanglaoDB", organism="human")

# Keep canonical cell type markers alone
#markers = markers[markers["canonical_marker"]]

# Remove duplicated entries
markers = markers[~markers.duplicated(["cell_type", "genesymbol"])]

#Format because dc only accepts cell_type and genesymbol

markers = markers.rename(columns={"cell_type": "source", "genesymbol":␣
 ↪"target"})
markers = markers[["source", "target"]]


markers.head()
```

[ ]:

	source	target
0	Hepatocytes	A1CF
1	Germ cells	A2APA5
2	Bergmann glia	A2M
3	Mast cells	A3FIN4
4	Mast cells	A4GALT

```
[ ]: #correct target to ensemble
markers = markers.merge(ensembl_var, left_on="target", right_on="gene_name",␣
 ↪how="left")
markers = markers.drop(columns=["target"])
# Remove duplicated entries
markers = markers[~markers.duplicated(["source", "ensembl_gene_id"])]

#Format because dc only accepts cell_type and genesymbol
markers = markers.rename(columns={"source": "source", "ensembl_gene_id":␣
 ↪"target"})

markers = markers[["source", "target"]]
markers = markers.dropna()

markers.head()
```

```
[ ]:          source           target
     0    Hepatocytes  ENSG00000148584
     2  Bergmann glia  ENSG00000175899
     4     Mast cells  ENSG00000128274
     6   Interneurons  ENSG00000115977
     7        Neurons  ENSG00000115977
```

```
[ ]: bm_mtx.var_names
```

```
[ ]: Index(['ENSG00000161920', 'ENSG00000122335', 'ENSG00000175548',
            'ENSG00000100330', 'ENSG00000176340', 'ENSG00000179846',
            'ENSG00000204860', 'ENSG00000172260', 'ENSG00000141424',
            'ENSG00000164512',
            …
            'ENSG00000164114', 'ENSG00000151702', 'ENSG00000224578',
            'ENSG00000138756', 'ENSG00000111052', 'ENSG00000176946',
            'ENSG00000150456', 'ENSG00000284934', 'ENSG00000261842',
            'ENSG00000260456'],
           dtype='object', length=17374)
```

```
[ ]: dc.mt.ulm(data=bm_mtx,
               net=markers,
               tmin = 3)
```

```
[ ]: score = dc.pp.get_obsm(bm_mtx, key="score_ulm")
```

```
[ ]: bm_mtx.obsm["score_ulm"].head(1)
```

```
[ ]:                         Acinar cells  Adipocyte progenitor cells  Adipocytes  \
     index
     Guo-AAACCTGAGAGCTTCT-2      0.883366                   -0.983141      0.4634
```

```
                        Adrenergic neurons  Airway goblet cells  Alpha cells  \
index
Guo-AAACCTGAGAGCTTCT-2           -0.592726            -1.026889     0.730972

                        Alveolar macrophages  Anterior pituitary gland cells  \
index
Guo-AAACCTGAGAGCTTCT-2              5.187966                       -0.233574

                        Astrocytes   B cells  …  Tanycytes  \
index                                         …
Guo-AAACCTGAGAGCTTCT-2    0.244758  1.139676  …   0.544296

                        Taste receptor cells  Thymocytes  Transient cells  \
index
Guo-AAACCTGAGAGCTTCT-2              0.291316   -0.543441         0.495932

                        Trigeminal neurons  Trophoblast cells  Tuft cells  \
index
Guo-AAACCTGAGAGCTTCT-2           -0.939089          -0.363228     3.11394

                        Undefined placental cells  Urothelial cells  \
index
Guo-AAACCTGAGAGCTTCT-2                  -0.784177         -0.784177

                        Vascular smooth muscle cells
index
Guo-AAACCTGAGAGCTTCT-2                       -0.5133

[1 rows x 163 columns]
```

[ ]:
```
bm_mtx.obsm["score_ulm"].columns
```

[ ]:
```
Index(['Acinar cells', 'Adipocyte progenitor cells', 'Adipocytes',
       'Adrenergic neurons', 'Airway goblet cells', 'Alpha cells',
       'Alveolar macrophages', 'Anterior pituitary gland cells', 'Astrocytes',
       'B cells',
       …
       'Tanycytes', 'Taste receptor cells', 'Thymocytes', 'Transient cells',
       'Trigeminal neurons', 'Trophoblast cells', 'Tuft cells',
       'Undefined placental cells', 'Urothelial cells',
       'Vascular smooth muscle cells'],
      dtype='object', length=163)
```

[ ]:
```
#rank genes
bm_gene_rank = dc.tl.rankby_group(score, groupby="leiden_res_",␣
 ↪reference="rest", method="t-test_overestim_var")
```

```
bm_gene_rank = bm_gene_rank[bm_gene_rank["stat"] > 0]
bm_gene_rank.head(5)
```

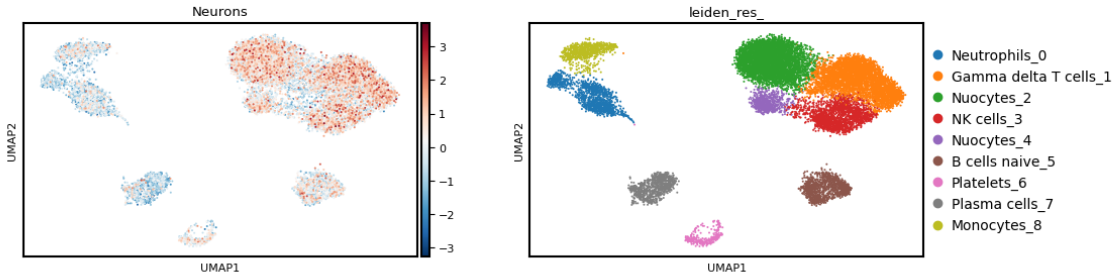
[ ]:
```
            group reference                 name        stat  meanchange  \
0  Neutrophils_0      rest          Neutrophils  113.123287    6.402901
1  Neutrophils_0      rest  Alveolar macrophages  106.999741    4.981299
2  Neutrophils_0      rest            Monocytes  101.253055    9.876754
3  Neutrophils_0      rest        Kupffer cells   89.814566    3.643749
4  Neutrophils_0      rest          Macrophages   89.083095    5.762394

   pval  padj
0   0.0   0.0
1   0.0   0.0
2   0.0   0.0
3   0.0   0.0
4   0.0   0.0
```

[ ]:
```
top_names_per_group = bm_gene_rank.groupby('group')['name'].apply(lambda x: x.
 ↪head(1))
display(top_names_per_group)
```

```
/tmp/ipython-input-2279155867.py:1: FutureWarning: The default of observed=False
is deprecated and will be changed to True in a future version of pandas. Pass
observed=False to retain current behavior or observed=True to adopt the future
default and silence this warning.
  top_names_per_group = bm_gene_rank.groupby('group')['name'].apply(lambda x:
x.head(1))
```

```
group
B cells naive_5        815           B cells naive
Gamma delta T cells_1  163     Gamma delta T cells
Monocytes_8            1304              Monocytes
NK cells_3             489                NK cells
Neutrophils_0          0               Neutrophils
Nuocytes_2             326          T memory cells
Nuocytes_4             653                Nuocytes
Plasma cells_7         1141           Plasma cells
Platelets_6            978               Platelets
Name: name, dtype: category
Categories (163, object): ['Acinar cells', 'Adipocyte progenitor cells',␣
 ↪'Adipocytes',
                           'Adrenergic neurons', …, 'Tuft cells', 'Undefined␣
 ↪placental cells',
                           'Urothelial cells', 'Vascular smooth muscle cells']
```

[ ]:
```
sc.pl.umap(score, color=["Neurons","leiden_res_"], cmap="RdBu_r")
```



[ ]:
```
n_ctypes = 3
ctypes_dict = bm_gene_rank.groupby("group").head(n_ctypes).
↪groupby("group")["name"].apply(lambda x: list(x)).to_dict()
ctypes_dict
```

```
/tmp/ipython-input-1047505464.py:2: FutureWarning: The default of observed=False
is deprecated and will be changed to True in a future version of pandas. Pass
observed=False to retain current behavior or observed=True to adopt the future
default and silence this warning.
  ctypes_dict = bm_gene_rank.groupby("group").head(n_ctypes).groupby("group")["n
ame"].apply(lambda x: list(x)).to_dict()
```

[ ]:
```
{'B cells naive_5': ['B cells naive', 'B cells memory', 'B cells'],
 'Gamma delta T cells_1': ['Gamma delta T cells',
  'NK cells',
  'Decidual cells'],
 'Monocytes_8': ['Monocytes', 'Dendritic cells', 'Macrophages'],
 'NK cells_3': ['NK cells', 'Gamma delta T cells', 'Decidual cells'],
 'Neutrophils_0': ['Neutrophils', 'Alveolar macrophages', 'Monocytes'],
 'Nuocytes_2': ['T memory cells', 'T cells naive', 'Nuocytes'],
 'Nuocytes_4': ['Nuocytes', 'T memory cells', 'Pancreatic progenitor cells'],
 'Plasma cells_7': ['Plasma cells', 'B cells', 'Goblet cells'],
 'Platelets_6': ['Platelets', 'Endothelial cells', 'Megakaryocytes']}
```

[ ]:
```
dict_ann = bm_gene_rank[bm_gene_rank["stat"] > 0].groupby("group").head(1).
↪set_index("group")["name"].to_dict()
dict_ann
```

```
/tmp/ipython-input-4239545895.py:1: FutureWarning: The default of observed=False
is deprecated and will be changed to True in a future version of pandas. Pass
observed=False to retain current behavior or observed=True to adopt the future
default and silence this warning.
  dict_ann = bm_gene_rank[bm_gene_rank["stat"] >
0].groupby("group").head(1).set_index("group")["name"].to_dict()
```

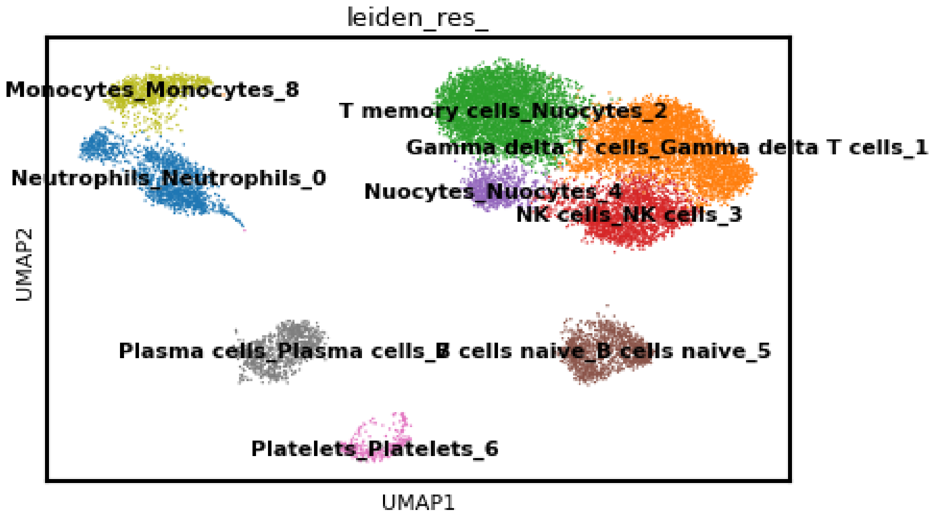
[ ]: {'Neutrophils_0': 'Neutrophils',
 'Gamma delta T cells_1': 'Gamma delta T cells',
 'Nuocytes_2': 'T memory cells',
 'NK cells_3': 'NK cells',
 'Nuocytes_4': 'Nuocytes',
 'B cells naive_5': 'B cells naive',
 'Platelets_6': 'Platelets',
 'Plasma cells_7': 'Plasma cells',
 'Monocytes_8': 'Monocytes'}

[ ]:
```
dict_ann_unique = {k: v + '_' + str(k) for k, v in dict_ann.items()}
display(dict_ann_unique)
```

```
{'Neutrophils_0': 'Neutrophils_Neutrophils_0',
 'Gamma delta T cells_1': 'Gamma delta T cells_Gamma delta T cells_1',
 'Nuocytes_2': 'T memory cells_Nuocytes_2',
 'NK cells_3': 'NK cells_NK cells_3',
 'Nuocytes_4': 'Nuocytes_Nuocytes_4',
 'B cells naive_5': 'B cells naive_B cells naive_5',
 'Platelets_6': 'Platelets_Platelets_6',
 'Plasma cells_7': 'Plasma cells_Plasma cells_7',
 'Monocytes_8': 'Monocytes_Monocytes_8'}
```

[ ]:
```
bm_mtx.obs["leiden_res_"] = bm_mtx.obs["leiden_res_"].cat.
↪rename_categories(dict_ann_unique)
```

[ ]:
```
plt.rcParams["figure.figsize"] = (5,3)  # Adjust figure size
# Reduce the default font size for text elements in plots
plt.rcParams['font.size'] = 8
plt.rcParams['axes.labelsize'] = 8
plt.rcParams['xtick.labelsize'] = 8
plt.rcParams['ytick.labelsize'] = 8

# Now replot the UMAP
sc.pl.umap(
    adata=bm_mtx,
    color=[ "leiden_res_"],#, 'cell_type'],
    ncols=2,
    legend_loc = 'on data',
    size=4,
)
```

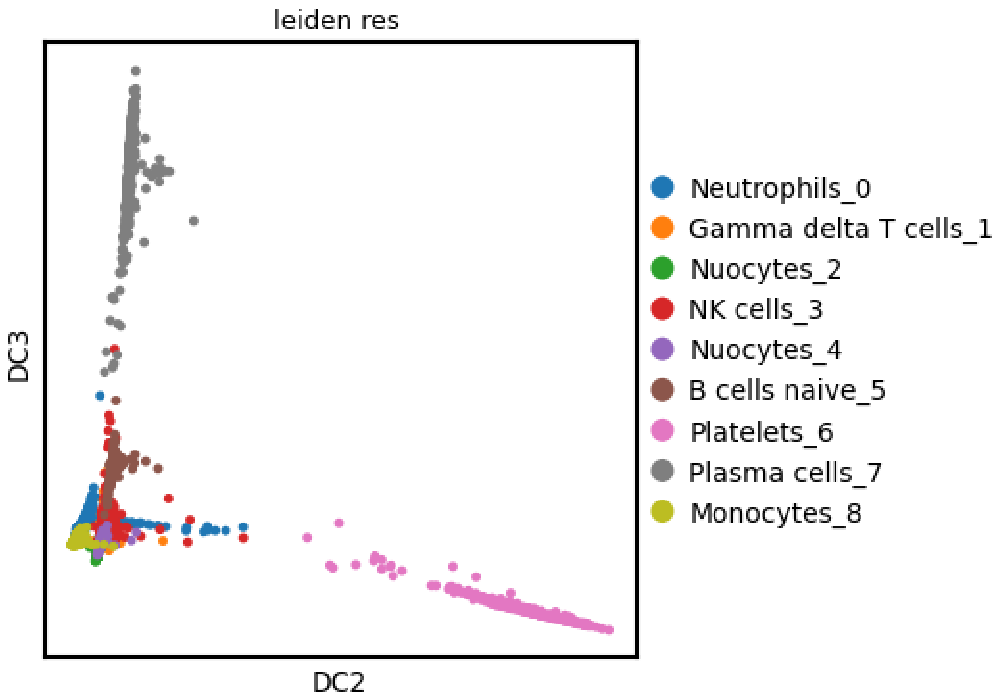


[ ]:

```python
#Pseudotime inference
sc.tl.diffmap(bm_mtx)
```

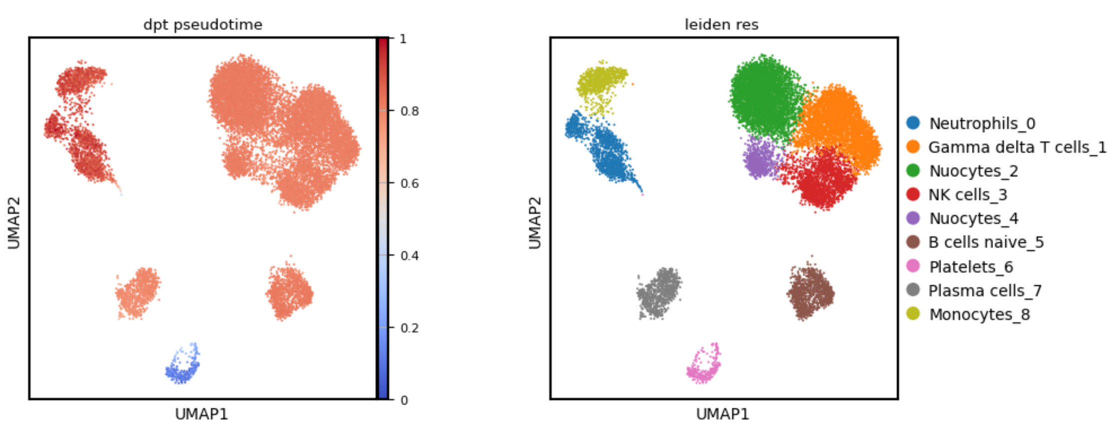
[ ]:

```python
plt.rcParams["figure.figsize"] = (4,4)
plt.rcParams['axes.labelsize'] = 10

root_ixs = bm_mtx.obsm["X_diffmap"][:, 3].argmin()
sc.pl.scatter(
    bm_mtx,
    basis="diffmap",
    color=["leiden_res_"],
    components=[2, 3],
    size=50
)

bm_mtx.uns["iroot"] = root_ixs
```



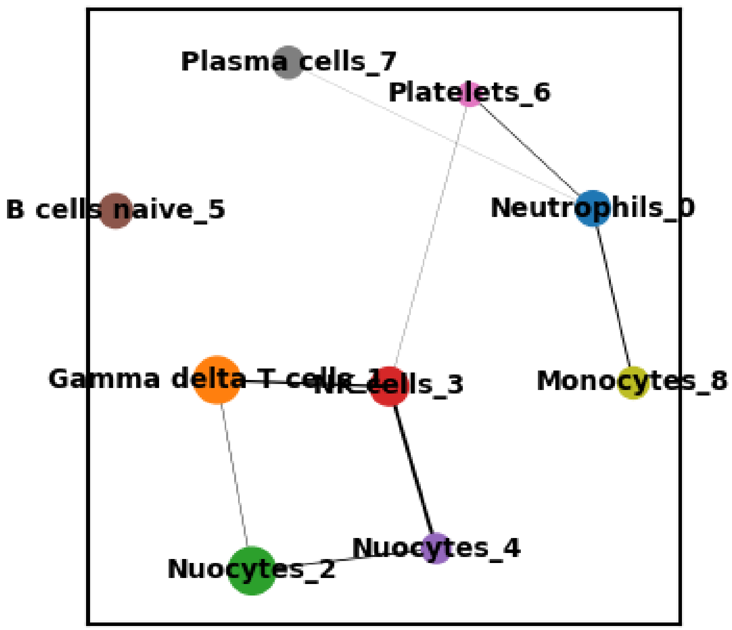
[ ]:
```
sc.tl.dpt(bm_mtx)
```

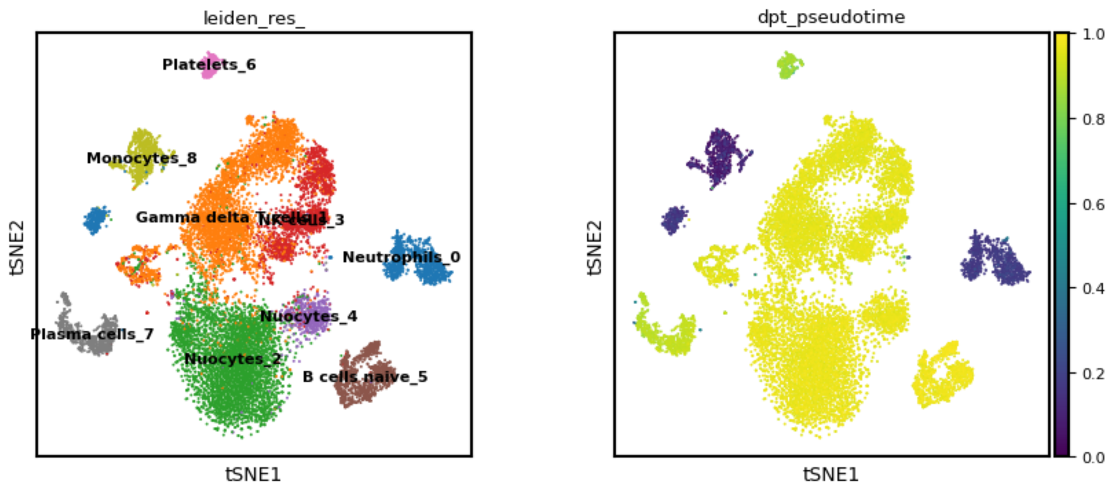
[ ]:
```
sc.pl.scatter(
    bm_mtx,
    basis="umap",
    color=["dpt_pseudotime", 'leiden_res_'],
    color_map="coolwarm",
)
```



[ ]:
```
#Trajectory analysis
root_celltype = "Monocytes_8"


sc.pp.neighbors(bm_mtx, use_rep="X_diffmap")
sc.tl.paga(bm_mtx, groups="leiden_res_")
iroot = np.flatnonzero(bm_mtx.obs["leiden_res_"] == root_celltype)[0]
bm_mtx.uns["iroot"] = iroot
sc.tl.dpt(bm_mtx)
```

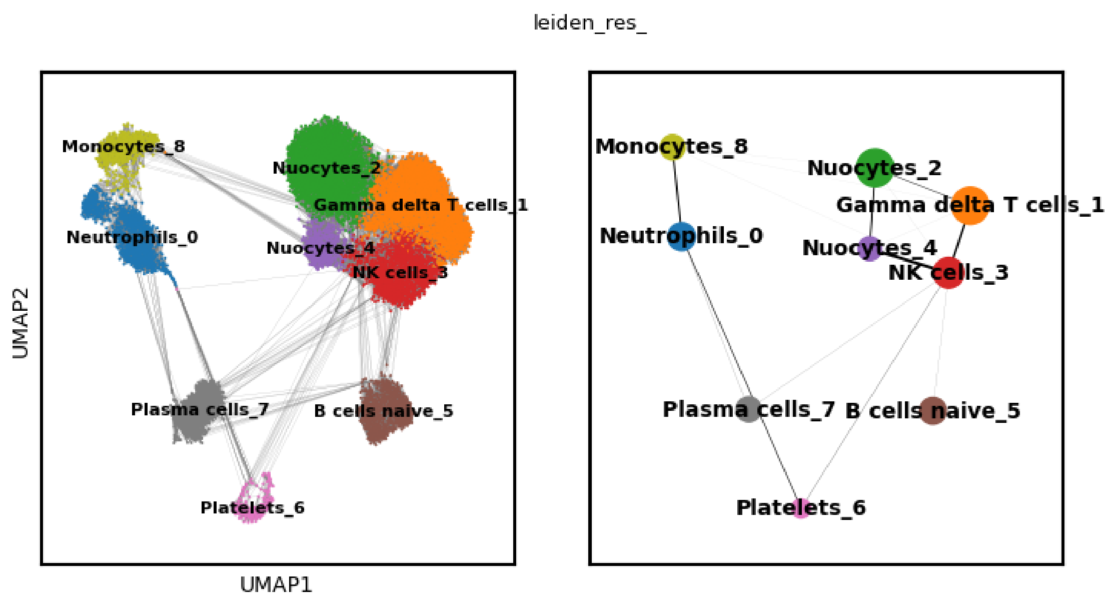
[ ]:
```
sc.pl.paga(bm_mtx, color=["leiden_res_"])
sc.pl.tsne(bm_mtx, color=["leiden_res_", "dpt_pseudotime"], legend_loc="on␣
↪data")
```





[ ]:

[ ]: `sc.pl.paga_compare(bm_mtx, threshold=0.001, frameon=True, edges=True)`



[ ]: [<Axes: xlabel='UMAP1', ylabel='UMAP2'>, <Axes: >]

[ ]: `sc.pl.draw_graph(bm_mtx, color=['dpt_pseudotime'], legend_loc='on data')`

[ ]: